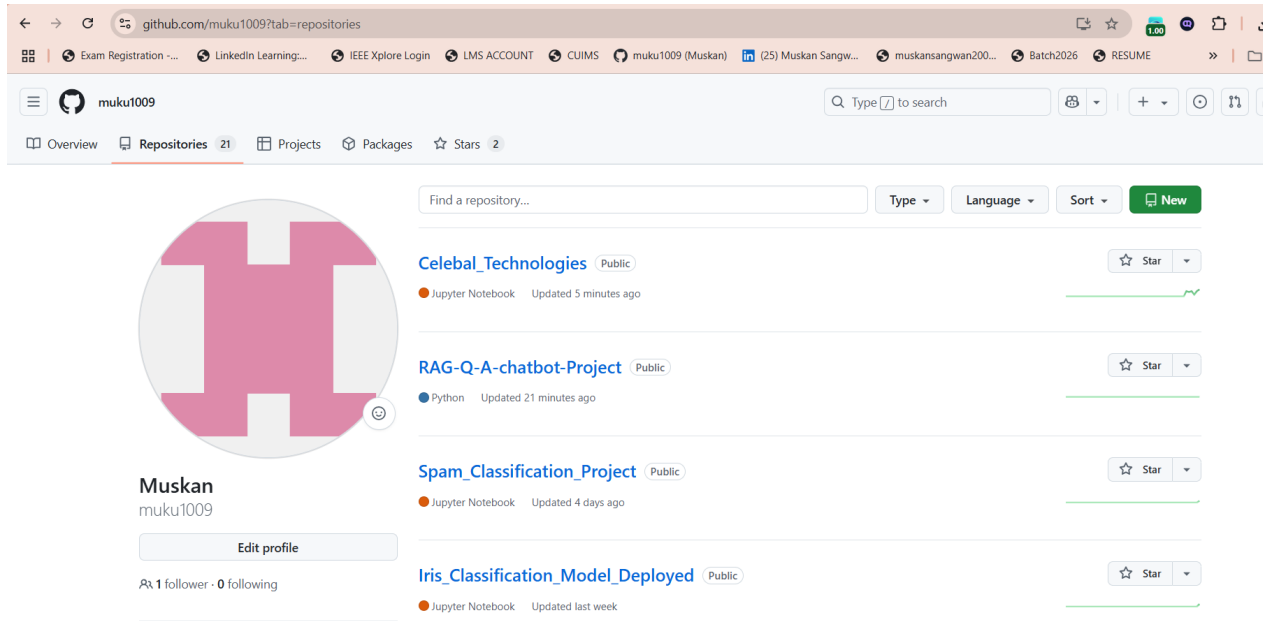


WEEK 8 : ASSIGNMENT

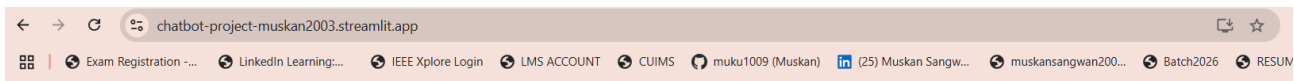
RAG-Based Loan Approval Chatbot

Project Links

- GitHub Repository: <https://github.com/muku1009/RAG-Q-A-chatbot-Project>



- Deployed Streamlit App: <https://chatbot-project-muskan2003.streamlit.app/>



Loan Approval Q&A Chatbot

 Ask questions about loan approvals. Powered by Retrieval-Augmented Generation (RAG) using FAISS & Gemini Pro.

 Ask a question:

e.g., Why was my loan rejected?

 Get Answer

Sample Questions You Can Try

-  What factors lead to loan rejection?
-  Tell me about applicants from rural areas.
-  Who mostly gets their loan approved?
-  Do self-employed people get loans easily?

 Developed as part of Celebal Technologies Internship – Week 8 Project

Project Overview

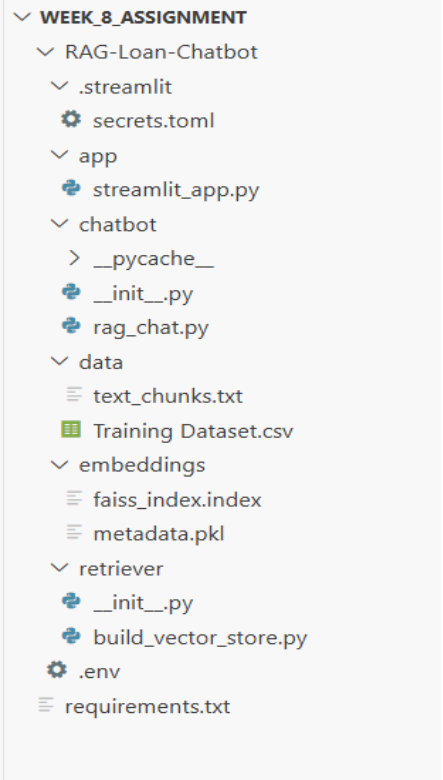
This project is a Loan Approval Q&A Chatbot developed using Retrieval-Augmented Generation (RAG) architecture. It allows users to ask natural language questions about historical loan application data and receive AI-generated responses using Google Gemini Pro and FAISS-based semantic search.

✅ Built as part of Week 8 Internship Assignment at Celebal Technologies.

Project Structure

RAG-Loan-Chatbot/

- └─ app/ → Streamlit app UI
- └─ chatbot/ → RAG logic using Gemini & FAISS
- └─ retriever/ → Data-to-vector store converter
- └─ data/ → Training data and chunked text
- └─ embeddings/ → FAISS index and metadata
- └─ .env → Local API key (not used on cloud)
- └─ .streamlit/secrets.toml → Deployment API key for Streamlit Cloud
- └─ requirements.txt → Python dependencies



```
WEEK_8_ASSIGNMENT
├── RAG-Loan-Chatbot
│   ├── .streamlit
│   │   └── secrets.toml
│   ├── app
│   │   └── streamlit_app.py
│   ├── chatbot
│   │   ├── __pycache__
│   │   ├── __init__.py
│   │   └── rag_chat.py
│   ├── data
│   │   ├── text_chunks.txt
│   │   └── Training Dataset.csv
│   ├── embeddings
│   │   ├── faiss_index.index
│   │   └── metadata.pkl
│   ├── retriever
│   │   ├── __init__.py
│   │   └── build_vector_store.py
│   ├── .env
│   └── requirements.txt
```

Technologies Used

- Python 3.10
- Streamlit

- Google Gemini Pro
- SentenceTransformers (all-MiniLM-L6-v2)
- FAISS
- dotenv
- tqdm, pandas, numpy

Workflow: End-to-End Pipeline

Step 1: Dataset Conversion & Embedding

File: retriever/build_vector_store.py

- Loads Training Dataset.csv
- Converts each loan entry into a human-readable paragraph
- Embeds text using SentenceTransformer
- Saves FAISS index and metadata

Step 2: Query Retrieval & Answer Generation

File: chatbot/rag_chat.py

- Loads vector index & metadata
- Finds top relevant chunks
- Uses Gemini to generate answer

Step 3: User Interface

File: app/streamlit_app.py

- Streamlit-based UI
- Sample questions, input box, clean styling

API Key Management

Environment	Key Storage	Access Code
Local	<code>.env</code> file	<code>os.getenv("GOOGLE_API_KEY")</code>
Streamlit Cloud	<code>secrets.toml</code>	<code>st.secrets["api_keys"]["GOOGLE_API_KEY"]</code>

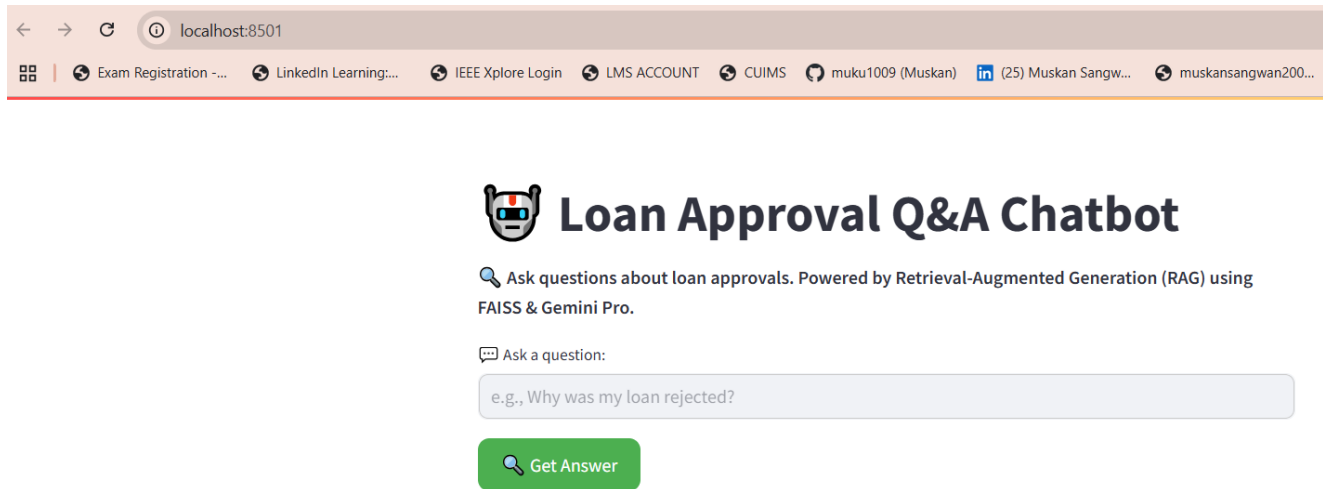
`.env` is ignored by Streamlit Cloud, hence `.streamlit/secrets.toml` is used during deployment.

Deployment Instructions

Local Run

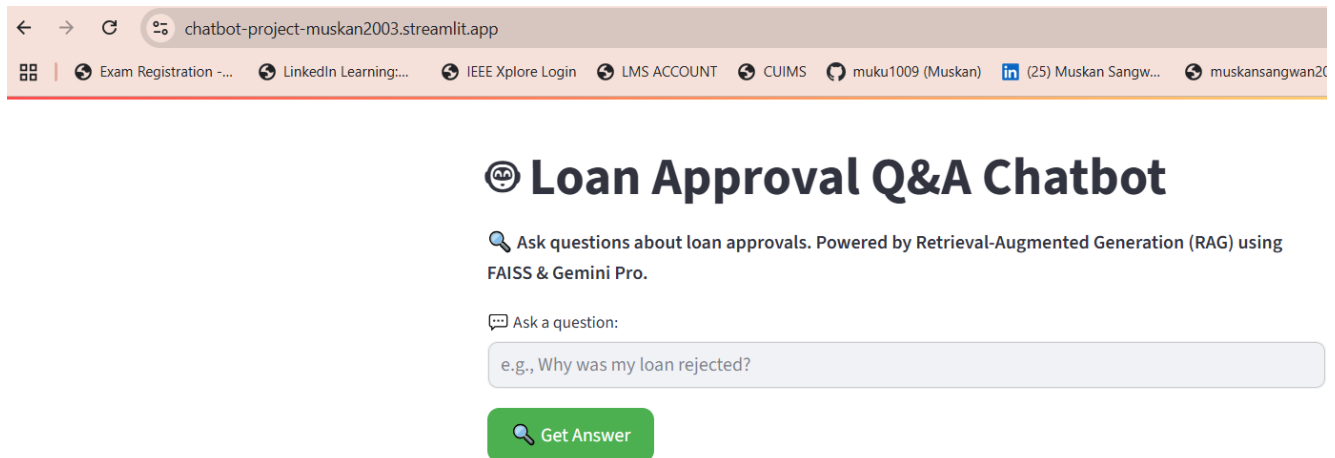
`pip install -r requirements.txt`

```
python retriever/build_vector_store.py
streamlit run app/streamlit_app.py
```



Streamlit Cloud

- Ensure requirements.txt is at root level
- Place API key inside `.streamlit/secrets.toml`
- Set app path as: RAG-Loan-Chatbot/app/streamlit_app.py



Features

- RAG-based Q&A chatbot
- Uses Gemini 1.5 for natural language generation
- FAISS for fast chunk retrieval
- Interactive and styled Streamlit UI
- Works both locally and on Streamlit Cloud

Sample Questions to Ask

- What are common reasons for loan rejection?
- Who mostly gets their loan approved?
- Do self-employed people get loans easily?
- Tell me about applicants from rural areas.

Loan Approval Q&A Chatbot

 Ask questions about loan approvals. Powered by Retrieval-Augmented Generation (RAG) using FAISS & Gemini Pro.

 Ask a question:

What factors lead to loan rejection?

 Get Answer

 Answer:

 Based on the provided data, all loan applications resulted in rejection and shared the common factor of a credit history that did not meet the guidelines. While other factors like income, loan amount, and property area varied, a failing credit history appears to be the primary reason for rejection in every case.
